

# Berry curvature and response functions from VASP wavefunctions

## VASPBERRY 1.3 scientific examples

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### Abstract

VASPBERRY evaluates geometric and response properties from VASP electronic states. This report illustrates the discrete-wavefunction and Kubo approaches with monolayer MoS<sub>2</sub>, a Bi bilayer and a three-septuple-layer MnBi<sub>2</sub>Te<sub>4</sub> film. The examples cover reciprocal-space Berry curvature, Chern and Z<sub>2</sub> indices, intrinsic charge Hall response, circular optical transitions and real-space spinor densities. Berry-curvature maps use Cartesian coordinates, band paths use distances along named high-symmetry directions, and the Z<sub>2</sub> n-field is shown in reduced coordinates to display its half-zone sums. The calculation guides provide the VASP files and executable commands.

## 1. Methods

### 1.1 Berry curvature from wavefunction overlaps

The Fukui–Hatsugai–Suzuki method evaluates the Berry phase around each cell of a periodic  $\mathbf{k}$  mesh using overlaps between neighboring wavefunctions. For an isolated group of bands, the overlap is a matrix and its determinant gives a gauge-invariant loop phase. With the connection  $\mathbf{A} = i\langle u | \nabla_{\mathbf{k}} u \rangle$  used here,

$$\Phi_p = -\arg(U_1(\mathbf{k})U_2(\mathbf{k} + \Delta\mathbf{k}_1)U_1(\mathbf{k} + \Delta\mathbf{k}_2)^{-1}U_2(\mathbf{k})^{-1}), \quad \overline{\Omega}_p = \frac{\Phi_p}{\Delta S_k}.$$

The normalized link  $U_j$  is the phase of the overlap determinant.  $\Delta S_k = |\mathbf{b}_1 \times \mathbf{b}_2|/(N_1 N_2)$  is the area of a mesh cell, with reciprocal vectors containing  $2\pi$ . Thus  $\Phi_p$  is a phase and  $\overline{\Omega}_p$  is an area-averaged curvature in  $\text{\AA}^2$ . The Chern number is  $C = (2\pi)^{-1} \sum_p \Phi_p$ . An isolated occupied group can include internal degeneracies; its separation from excluded bands remains essential. The discrete construction and its band-group extension are described by Fukui, Hatsugai and Suzuki.

The map and integral convey different information. A material can have strong local curvature at opposite valleys while its total Chern number is zero. An integer lattice Chern result also needs a sufficiently resolved mesh and a physically appropriate isolated band group.

## 1.2 Kubo curvature and Hall conductivity

For an isolated band and a Cartesian Hamiltonian derivative  $D_\alpha = \partial H / \partial k_\alpha$ ,

$$\Omega_{n,xy}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

This form resolves curvature at each sampled  $\mathbf{k}$  point. Sharp curvature near small gaps makes the  $\mathbf{k}$  sampling and intermediate-band window particularly important. Individual-band values are not used where the sampled bands cannot be resolved separately. The native WAVECAR implementation uses canonical momentum of the stored pseudo-wavefunctions; full material velocity can require PAW, nonlocal and SOC terms. The exported-matrix interface permits an explicitly supplied physical operator.

For an isolated occupied bundle  $\mathcal{V}$ , the trace curvature is

$$\Omega_{xy}^{\mathcal{V}} = -2 \text{Im} \sum_{n \in \mathcal{V}} \sum_{m \notin \mathcal{V}} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

Internal occupied-pair terms cancel in the sum of band curvatures. Excluding them before division avoids singular individual-band terms at internal degeneracies. The bundle must remain separated from the excluded bands. This occupied-to-unoccupied formulation and its numerical advantage are given by Wang et al., Eq. (11) and Sec. III D.

For a two-dimensional system, the intrinsic charge sheet conductivity is

$$\sigma_{xy}(\mu, T) = -\frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} f(E_{n\mathbf{k}} - \mu, T) \Omega_{n,xy}(\mathbf{k}).$$

Within an insulating gap at zero temperature, this becomes  $\sigma_{xy} = -(e^2/h)C_{\text{occ}}$  in the stated convention. General background on Berry curvature and electronic transport is given in Xiao, Chang and Niu. The Bi Hall example evaluates the occupied-group flux directly; it does not assign independent curvatures to unresolved Kramers partners.

For chemical potentials crossing band edges, the Kubo implementation uses unordered interband pairs. With  $N_{nm}^{xy} = -2 \text{Im}(D_{x,nm} D_{y,mn})$ , the occupation-weighted integrand is  $\sum_{n < m} (f_n - f_m) N_{nm}^{xy} / (E_n - E_m)^2$ . Equal-occupation pairs cancel before division. Changes from a gap reference are evaluated with occupation differences before summation. This preserves small doping responses without subtracting large filled-band baselines. The temperature enters only the Fermi function; the electronic structure remains fixed. Regional contributions integrate the same charge response over specified parts of the BZ, with a K-K' difference defined without a factor of one-half. Such a partition is not a separately conserved valley-current operator.

### 1.3 PAW optical matrices and full-connection interpolation

For an insulating occupied bundle, the supported standard VASP longitudinal optical calculation supplies matrix elements  $C_{cv,\alpha} = \langle u_c | \partial_{k_\alpha} u_v \rangle$  in Å. Here  $v$  and  $c$  label occupied and empty states. Their contribution to the trace is

$$\Omega_{xy}^{\text{occ}} = -2 \text{Im} \sum_{v,c} C_{cv,x}^* C_{cv,y}.$$

The energy denominator is already contained in these derivatives. The longitudinal PAW optical expression includes projector and augmentation terms, as described by Gajdoš et al.. The waver-hall route checks the actual VASP output, occupied filling and global gap before integration. Its present scope is VASP 5.4.4, zero temperature and a fixed insulating bundle; the occupied–empty separation must exceed the producer’s 2 meV degeneracy threshold. Mesh and empty-state convergence remain separate requirements.

For narrow curvature peaks, VASPBERRY also interpolates the full Hamiltonian and position matrices of a VASP-derived Wannier model. Its own Fourier, diagonalization and occupied-trace routines evaluate  $\Omega = J0+J1+J2$ : the basis-connection curl, the mixed connection/Hamiltonian-derivative term, and the derivative-pair term. Only occupied-to-empty denominators occur; internal degeneracies are allowed. The formulation follows Wang et al. and Lopez et al., Eq. (51). Bands and local curvature are checked against direct VASP data. The separate postw90 calculation in Wannier90 provides an independent check of VASPBERRY’s evaluation of the same finite model.

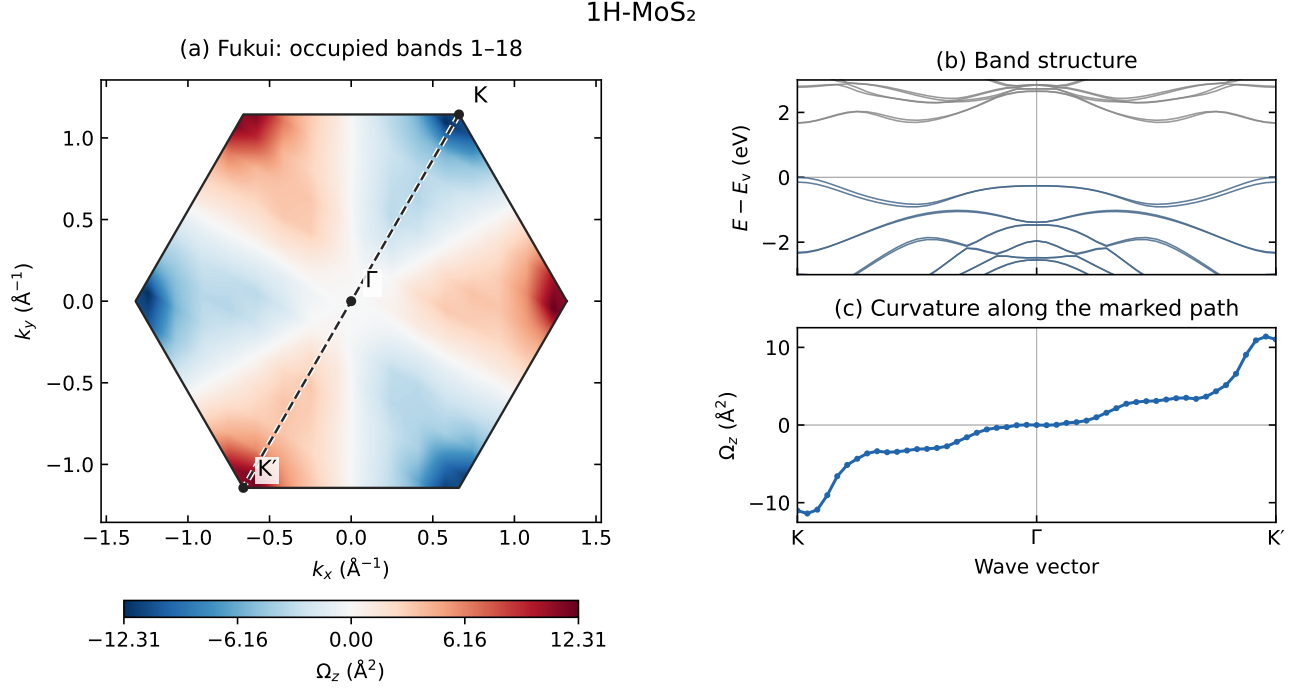
## 2. Materials and numerical settings

| Dataset                                                   | Wavefunctions and sampling                                                             | Quantities shown                                                        |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Monolayer MoS <sub>2</sub> , full zone                    | SOC; 12×12 mesh; 26 bands; 400 eV                                                      | Fukui and Kubo occupied-bundle curvature, bands 1–18                    |
| Monolayer MoS <sub>2</sub> , matching path                | SOC; 49 K–Γ–K’ points; 26 bands; 400 eV                                                | Band structure and Kubo curvature beside the maps                       |
| Monolayer MoS <sub>2</sub> , local valleys                | SOC; two 9×9 patches; 26 bands; 400 eV                                                 | Isolated band-18 curvature near K and K’                                |
| Monolayer MoS <sub>2</sub> , transport                    | SOC; 12×12 to 36×36 meshes; stored and retained band windows varied separately; 400 eV | Chemical-potential and temperature dependence of regional Hall response |
| Monolayer MoS <sub>2</sub> , supplied path                | SOC; 48 K–Γ–K’ points; 32 bands; 400 eV                                                | Optical spectra and Γ-point state density                               |
| Buckled Bi bilayer                                        | SOC; full 12×12 mesh; 18 bands; occupied bands 1–10                                    | Fukui–Hatsugai n-field and Z <sub>2</sub> invariant                     |
| MnBi <sub>2</sub> Te <sub>4</sub> , three septuple layers | SOC+U; 21 atoms; full 6×6 VASP mesh; 192 bands; occupied bands 1–123; 270 eV           | Magnetic Chern invariant, bands and Hall-integration checks             |

The input guide links the structures and VASP files. The MoS<sub>2</sub> maps and their matching path use the same public SCF charge density, structure, potentials, cutoff and 26-band window. VASP 5.4.4 generated the 12×12 mesh and 49-point path with ICHARG=11 and ISYM=–1; only the k-point list changes. The sampled occupied-to-empty direct gap is 1.674 eV. The tutorial supplies preparation files and commands for both WAVECARS. The optical and real-space examples retain the separate supplied 32-band path.

### 3. Results

#### 3.1 MoS<sub>2</sub>: Fukui Berry curvature over the full Brillouin zone

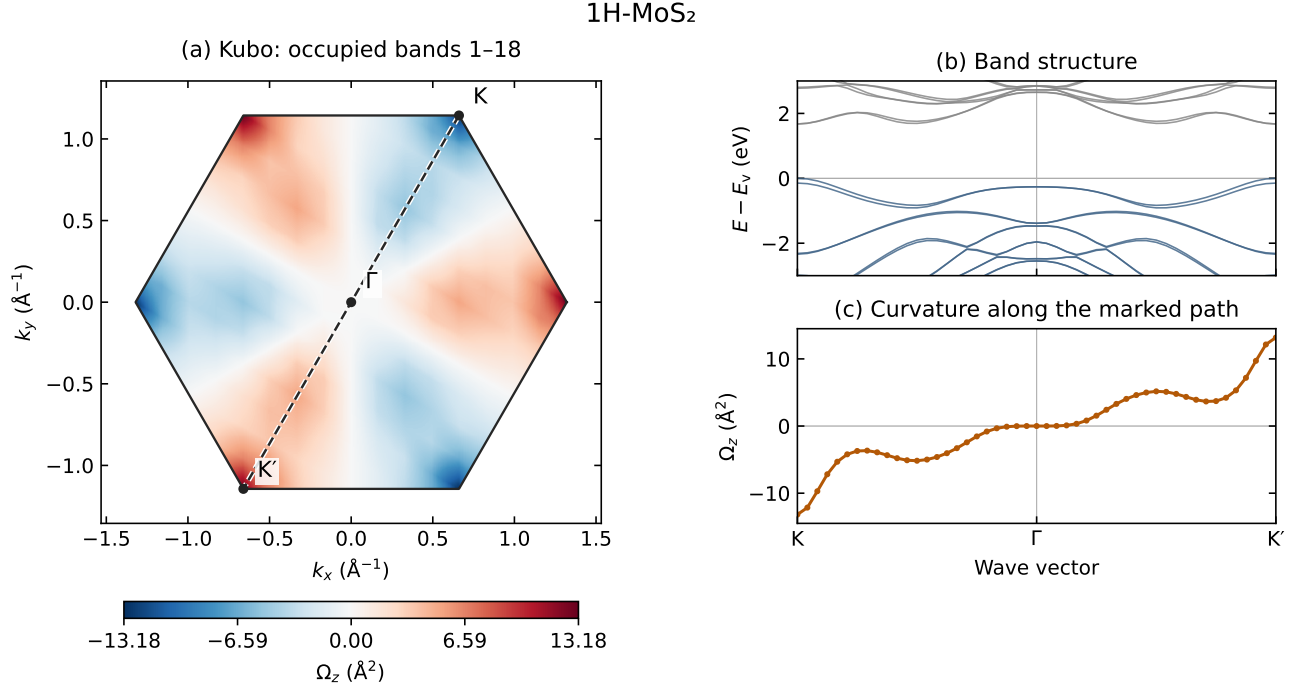


**Figure 1.** (a) Fukui Berry curvature of occupied bands 1–18 in the Cartesian first Brillouin zone. The dashed line marks K– $\Gamma$ –K', with  $K = (1/3, 2/3)$  and  $K' = -K$  in reciprocal coordinates. Native  $12 \times 12$  plaquette values are displayed with periodic bilinear interpolation onto a  $401 \times 401$  Cartesian grid. (b) Band structure along that path; energies are relative to the valence-band maximum, with occupied bands in blue and empty bands in gray. (c) A periodic bilinear cut of the plaquette field along the marked path. This line visualizes panel (a), rather than adding an independent pointwise curvature calculation. Both right-hand panels share the same Cartesian path distance.

The curvature has opposite signs near the time-reversed K and K' valleys. The largest sampled magnitudes are approximately **12.31** Å<sup>2</sup>, while the full-zone Chern number is zero to numerical precision. The native text map integrates to approximately  $-5 \times 10^{-7}$  after decimal rounding. This illustrates the distinction between valley-contrasting local geometry and vanishing total charge Chern number. The connection between inversion breaking, spin-orbit coupling and valley optical response in MoS<sub>2</sub> is discussed by Xiao et al.. The present map is a finite-mesh example rather than a convergence study.

Calculation and plotting commands

### 3.2 MoS<sub>2</sub>: Kubo curvature of the occupied valence-band bundle



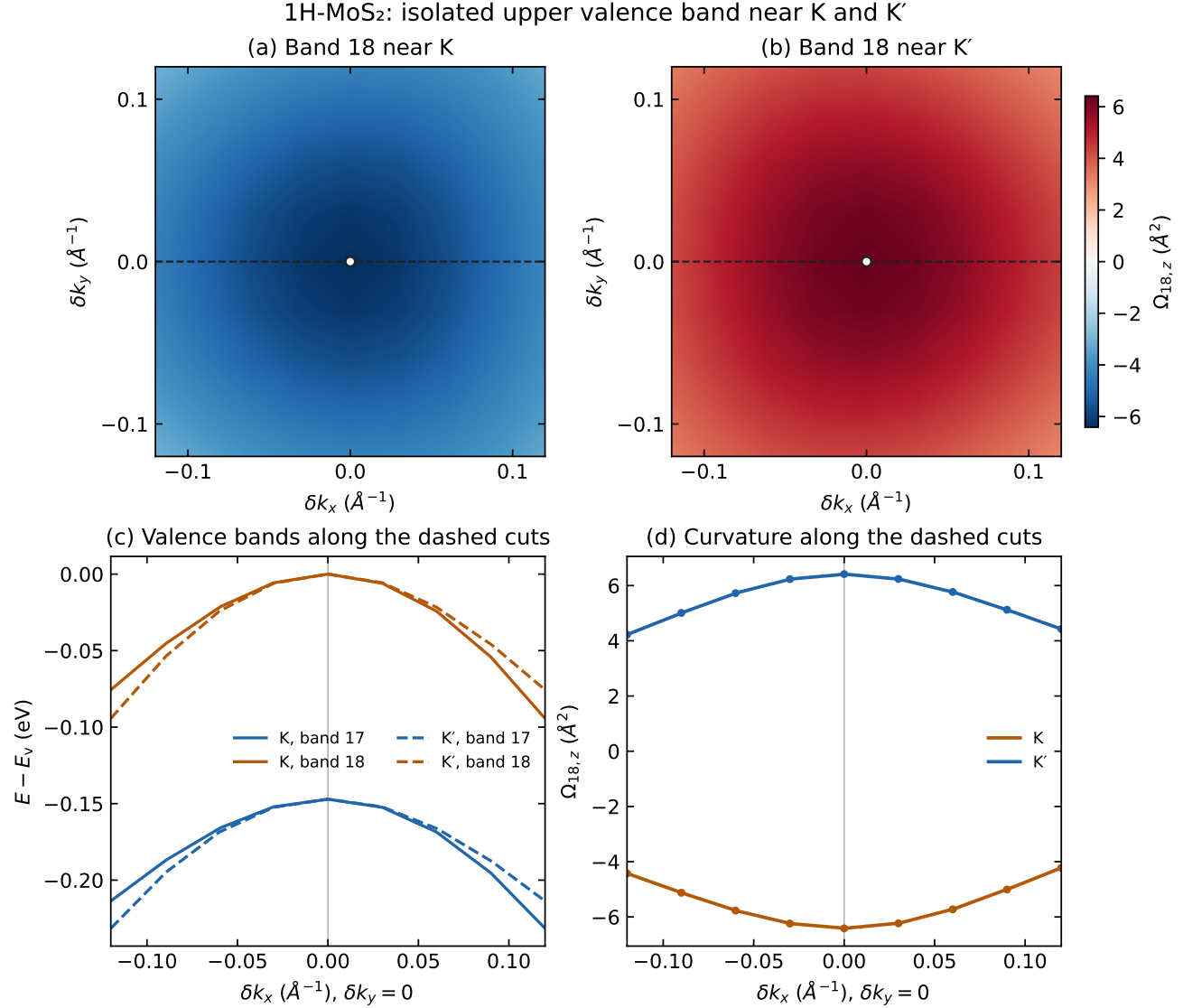
**Figure 2.** (a) Canonical-momentum Kubo trace curvature of occupied bands 1–18 on the  $12 \times 12$  mesh. Only transitions to the stored empty bands 19–26 enter the sum. The map uses the same periodic bilinear display interpolation and dashed  $K$ – $\Gamma$ – $K'$  path as Figure 1. (b) Band structure, with occupied bands in blue and empty bands in gray. (c) Bundle curvature calculated directly at all 49 path points. Internal valence-band degeneracies do not interrupt the bundle curve.

The occupied-bundle curvature is approximately  $-13.18 \text{ \AA}^2$  at  $K$  and  $+13.18 \text{ \AA}^2$  at  $K'$ . The occupied space stays separated from the empty states by at least **1.674 eV** on both samplings, so every mesh and path point is valid for the bundle calculation. The native option `-kubo_bundle 1` performs the external-band sum directly, avoiding large cancelling internal terms.

Figures 1 and 2 now describe the same occupied band space. Their numerical values need not coincide at this resolution: Fukui gives finite-plaquette averages, while this Kubo result uses pointwise canonical momentum and a finite empty-band window. The  $12 \times 12$  mesh and 26 stored bands require convergence for quantitative predictions. NumPy interpolation smooths only the display; all integration and numerical checks use the original samples.

Calculation and interpretation

### 3.2.1 MoS<sub>2</sub>: an isolated single band near the valleys

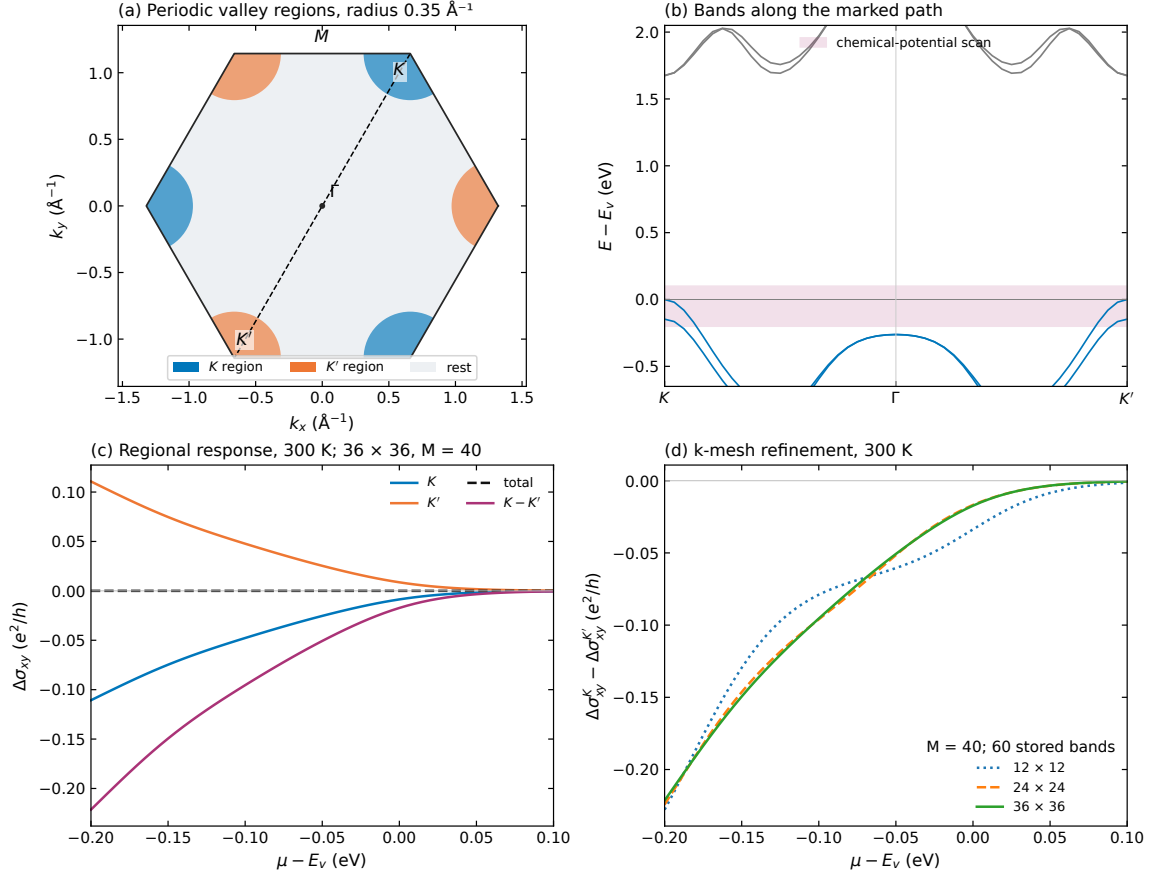


**Figure 3.** Single-band Kubo curvature of the upper valence band, band 18, around K and K'. Each local Cartesian patch extends  $\pm 0.12 \text{ \AA}^{-1}$  and contains  $9 \times 9$  VASP points. The upper maps use NumPy bilinear display interpolation; the dashed lines mark the cuts below. The lower panels show the two upper valence bands near both valleys and band-18 curvature at both valleys. Symbols mark the calculated curvature samples.

Band 18 touches its Kramers partner at  $\Gamma$ , but is separated from every other stored band by at least **0.130 eV** throughout these valley patches. The 17–18 splitting at K is **0.147 eV**, and the valley curvatures are approximately  **$-6.415/+6.415 \text{ \AA}^2$**  at **K/K'**. This is a suitable local single-band example of the spin-split valley physics described by Xiao et al.. The patch data describe local curvature; a full-zone integral requires the appropriate isolated band bundle and complete BZ sampling.

VASP preparation, calculation and reference data

### 3.2.2 MoS<sub>2</sub>: intrinsic charge and regional valley Hall response



**Figure 4.** (a) Cartesian first BZ with periodic K/K' disks of radius  $0.35 \text{ \AA}^{-1}$  and the marked K- $\Gamma$ -K' line. (b) The matching 26-band dispersion, with the chemical-potential interval shaded. (c) Regional Hall changes at 300 K on the  $36 \times 36$  mesh, using 60 stored bands and pairs within bands 1–40. (d) Mesh refinement at that fixed band window. Energies are relative to each source's valence-band maximum; the band path uses the same density and cutoff.

The 61-point scan covers  $E_v - 0.20$  to  $E_v + 0.10$  eV at 0 and 300 K. We define  $\Delta\sigma(\mu) = \sigma(\mu) - \sigma(\mu_{\text{ref}})$  with a midgap reference, and the valley difference as  $\Delta\sigma_K - \Delta\sigma_{K'}$ . The total charge response remains below  $2.24 \times 10^{-9} e^2/h$ , consistent with time-reversal symmetry. Numerical tables also give absolute conductivities and carrier counts.

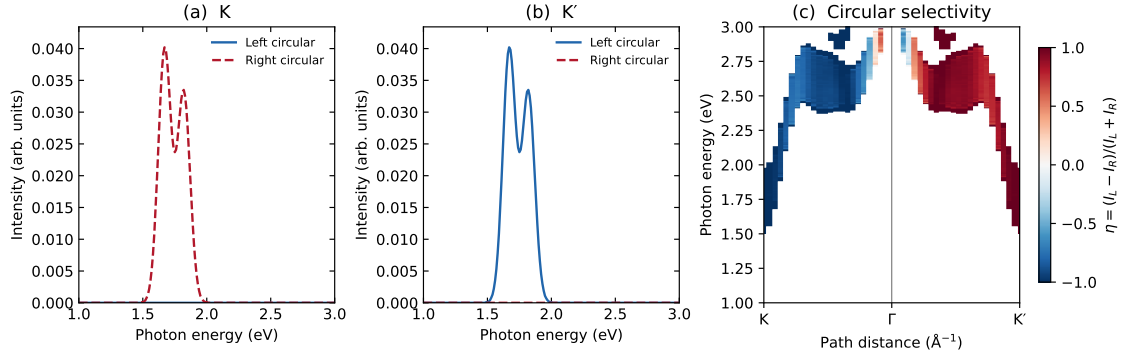
Two numerical choices were varied separately at 300 K. The relative L2 difference below is the norm of the curve change divided by the finer curve's norm, on a common  $\mu - E_v$  grid.

| Test                                             | Refinement                              | Relative L2 change |
|--------------------------------------------------|-----------------------------------------|--------------------|
| k mesh; 60 stored bands, pair cutoff 40          | $12 \times 12 \rightarrow 24 \times 24$ | 12.89%             |
| k mesh; same band window                         | $24 \times 24 \rightarrow 36 \times 36$ | 1.482%             |
| Pair cutoff; one $12 \times 12$ , 96-band source | $60 \rightarrow 80$                     | 1.263%             |
| Pair cutoff; same source                         | $80 \rightarrow 90$                     | 0.202%             |

The final mesh change exceeds the stated 1% criterion. The last cutoff increment is below 1%; these separate tests do not establish joint convergence. The reference guide includes the commands and additional checks on empty-state accuracy and numerical degeneracy grouping.

This rigid-band response uses canonical momentum, omitting PAW augmentation and nonlocal/SOC velocity terms. Numerical refinement does not remove those approximations. The supplementary plot retains the finite-mesh steps at 0 K.

### 3.3 MoS<sub>2</sub>: valley-selective circular transitions



**Figure 5.** Left- and right-circular spectra at K and K' and the selectivity  $\eta = (I_L - I_R)/(I_L + I_R)$  along the path. Normal incidence is used, with 0.05 eV Gaussian broadening and transitions from occupied bands 1–18 to bands 19–20. White regions in the selectivity map exclude negligible intensity.

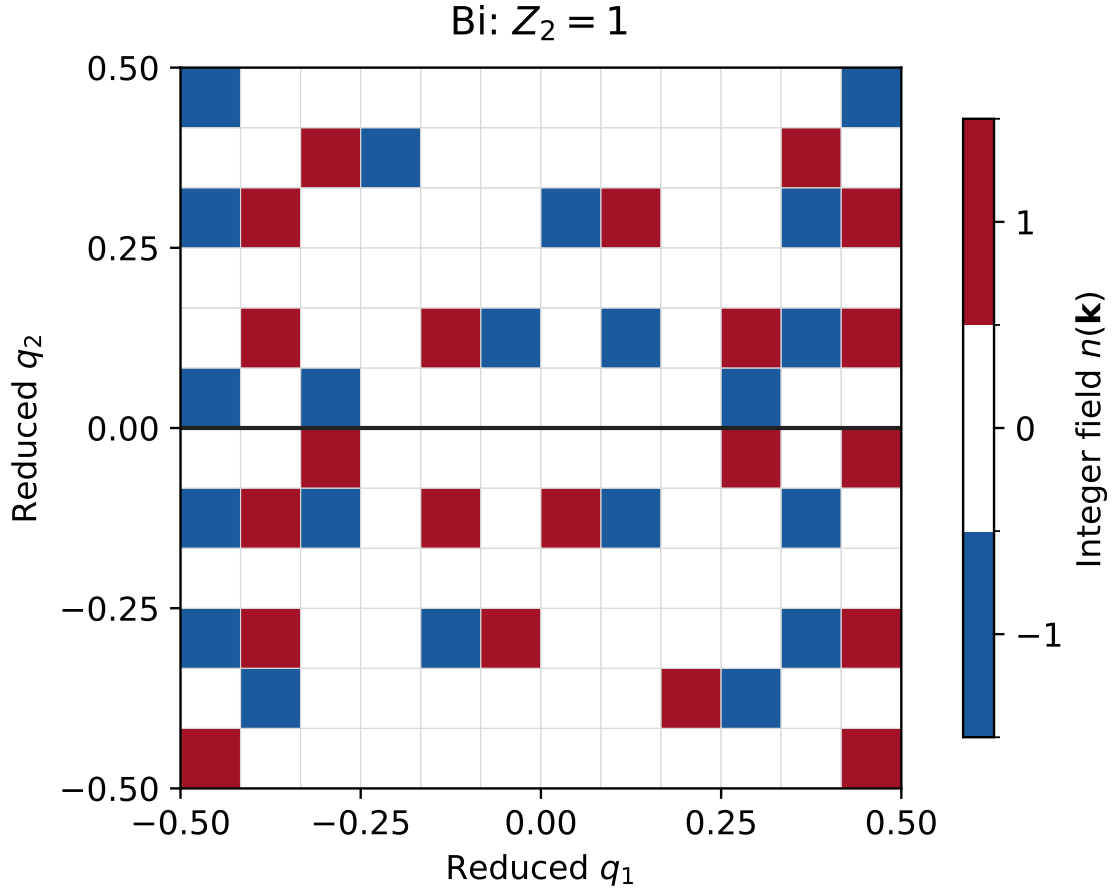
The first peak occurs near **1.67 eV**. Its selectivity is  $-1$  at K and  $+1$  at K' in VASPBERRY's channel convention. The opposite helicities provide a clear illustration of valley-selective transitions. Intensities are in arbitrary units and retain the native sampling normalization; an absolute absorption spectrum requires a full-zone calculation and the appropriate optical matrix elements.

Optical calculation guide

### 3.4 Bi: Fukui–Hatsugai $Z_2$ invariant and n-field map

For the Bi bilayer, VASPBERRY evaluates the Fukui–Hatsugai lattice n-field from the occupied SOC bands 1–10 on a full  $\Gamma$ -centered  $12 \times 12$  mesh. With the time-reversal-compatible gauge used in this construction, the  $Z_2$  invariant is the parity of the integer-field sum over a half Brillouin zone:

$$\nu = \left[ \sum_{p \in B_{1/2}} n(p) \right] \bmod 2.$$



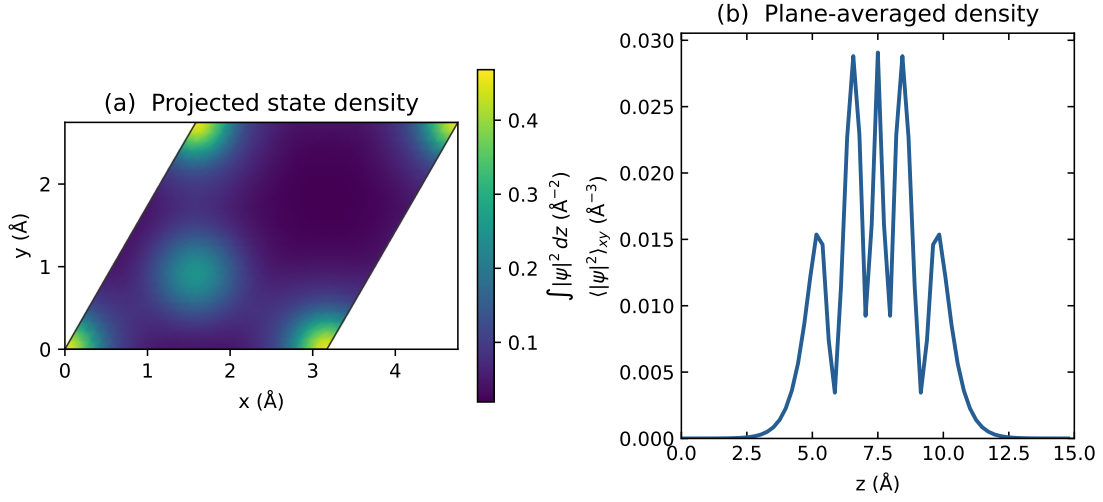
**Figure 6.** Fukui–Hatsugai integer field  $n(\mathbf{k})$  calculated from the Bi VASP spinor wavefunctions. Red, white and blue denote +1, 0 and -1, respectively. The native plaquettes are displayed without interpolation in dimensionless reduced coordinates,  $\mathbf{k} = q_1 \mathbf{b}_1 + q_2 \mathbf{b}_2$ , with opposite edges periodically identified. The line at  $q_2 = 0$  separates the upper and lower half zones. The occupied-band calculation gives  $Z_2 = 1$ .

The upper and lower half-zone sums are **-3 and +3**, respectively. Both are odd, so their parities agree at  $\nu = 1$ , identifying the nontrivial time-reversal-symmetric insulating phase for this calculation. The sampled minimum direct and global gaps are **0.592 eV** and **0.510 eV**.

The local n-field depends on the gauge and logarithm branch; its half-zone parity is the invariant. The reduced-coordinate map shows the original mesh half-zones used for these sums. The separate occupied  $\mathbf{C} = \mathbf{0}$  and zero charge-Hall response are consistent with this nontrivial  $Z_2$  result.

$Z_2$  calculation and plotting guide · Numerical n-field data

### 3.5 MoS<sub>2</sub>: a real-space $\Gamma$ state



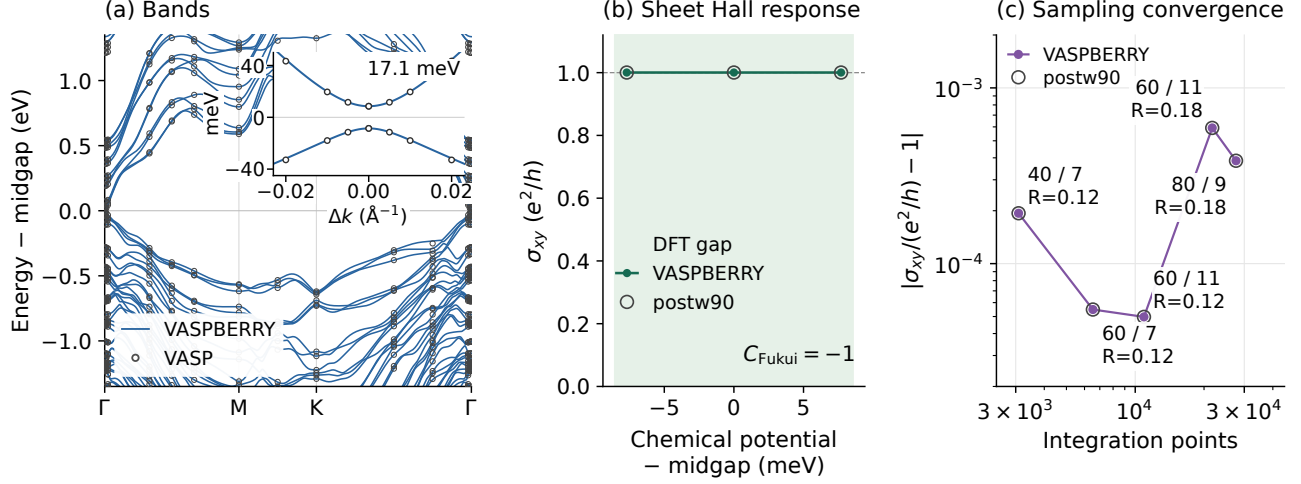
**Figure 7.** Projected pseudo-wavefunction density of band 18 at  $\Gamma$  and its plane average along  $z$ . The in-plane plot uses the Cartesian geometry of the oblique primitive cell. Both spinor components are retained.

The integrated pseudo-density is approximately **0.8258**, consistent with the plane-wave coefficient norm. The PAW augmentation is not part of this plot, so a unit density integral is not imposed. The example illustrates conversion of complex amplitudes into a physical-coordinate density map.

Wavefunction calculation guide

### 3.6 MnBi<sub>2</sub>Te<sub>4</sub>: Chern number and VASPBERRY Hall response

This example connects an occupied-band topological invariant to a Hall integral computed by VASPBERRY. The three-septuple-layer MnBi<sub>2</sub>Te<sub>4</sub> film has 21 atoms with +z/-z/+z Mn magnetization. Its fixed, unrelaxed geometry retains bulk vertical spacings from Yan et al., with in-plane  $a = 4.336$  Å, PBE+SOC and Mn  $U_{\text{eff}} = 5.34$  eV following Otrokov et al. It is distinct from the relaxed films in that study.



**Figure 8.** (a) VASPBERRY-interpolated bands along  $\Gamma$ -M-K- $\Gamma$ , with direct VASP checks; the inset resolves the K- $\Gamma$ -M gap near  $\Gamma$ . (b) VASPBERRY full-connection sheet Hall response at three chemical potentials inside the gap, at zero temperature; open markers show the independent postw90 comparison. The dashed line is the topological expectation. (c) Deviation  $|\sigma_{xy}/(e^2/h) - 1|$  under integration refinement. Labels give the base mesh, local subdivision and  $\Gamma$ -region radius in  $\text{\AA}^{-1}$ . Both solvers use the same model and weights; their agreement is distinct from mesh convergence.

The VASP gap is **17.10 meV**. Fukui links of the **complete occupied valence-band bundle, bands 1–123, give  $C = -1$** , corresponding to  $\sigma_{xy} = +e^2/h$  in our convention. A direct  $6 \times 6$  PAW optical integral instead gives **163.109  $e^2/h$** : the  $\Gamma$ -point curvature reaches approximately  $-1.52 \times 10^4 \text{ \AA}^2$ , and this coarse mesh overweights the narrow peak. An integer Fukui result does not establish convergence of the pointwise integral.

For dense integration, VASPBERRY’s wannier-hall evaluates all J0/J1/J2 terms from the supplied VASP-derived Hamiltonian and position matrices. wannier-bands independently generates the plotted dispersion. The model contains 138 orbitals and **87 occupied states**; the omitted 36 deep VASP bands form a separately checked  $C = 0$  bundle. Near  $\Gamma$ , direct VASP band edges agree within 1.26 meV and local curvature within 1.44%; the latter comparison retains the full-123 versus model-87 subspace difference. The maximum band-edge difference across all twelve validation locations is 18.14 meV.

VASPBERRY’s final 27,600-point result is  $\sigma_{xy} = 1.000384978 e^2/h$  at all three gap chemical potentials. It uses an  $80 \times 80$  base mesh and  $9 \times 9$  subdivisions in the  $0.18 \text{ \AA}^{-1}$   $\Gamma$  region. The last refinement changes the result by  $2.07 \times 10^{-4} e^2/h$ , below the stated  $10^{-3} e^2/h$  comparison criterion. The nonmonotonic region-expansion control is retained; no integer rounding is applied. Independent postw90 totals and all three component decompositions agree within their original output precision. A separate  $160 \times 160$  energy scan places all three chemical potentials inside the sampled model gap.

This validates VASPBERRY’s full-connection evaluation of a fixed finite model. The source  $6 \times 6$  mesh, structure and basis still require convergence for quantitative material predictions; localization stopped before its spread tolerance. The material tutorial provides the actual inputs, VASPBERRY commands and outputs. The independent external reference is retained separately. This input route requires both Wannier operators; it does not infer missing PAW/nonlocal/SOC velocity terms from a WAVECAR alone.

## 4. Practical use

Begin with the example guide, reproduce the chosen quantity, and then select the sampling and band window appropriate to your own material. Use Cartesian reciprocal-space maps to display lattice symmetry, named k paths for band-resolved quantities, and energy or chemical potential for spectra and transport. State units, represented bands, temperature and broadening in the caption.

For quantitative conclusions, converge the VASP electronic structure, k mesh and relevant band window. The examples show how the methods are used; the supplementary analytic references check formulas and conventions, while validation details document the numerical implementation. Detailed execution records remain alongside the numerical files for reproducibility.

## References

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